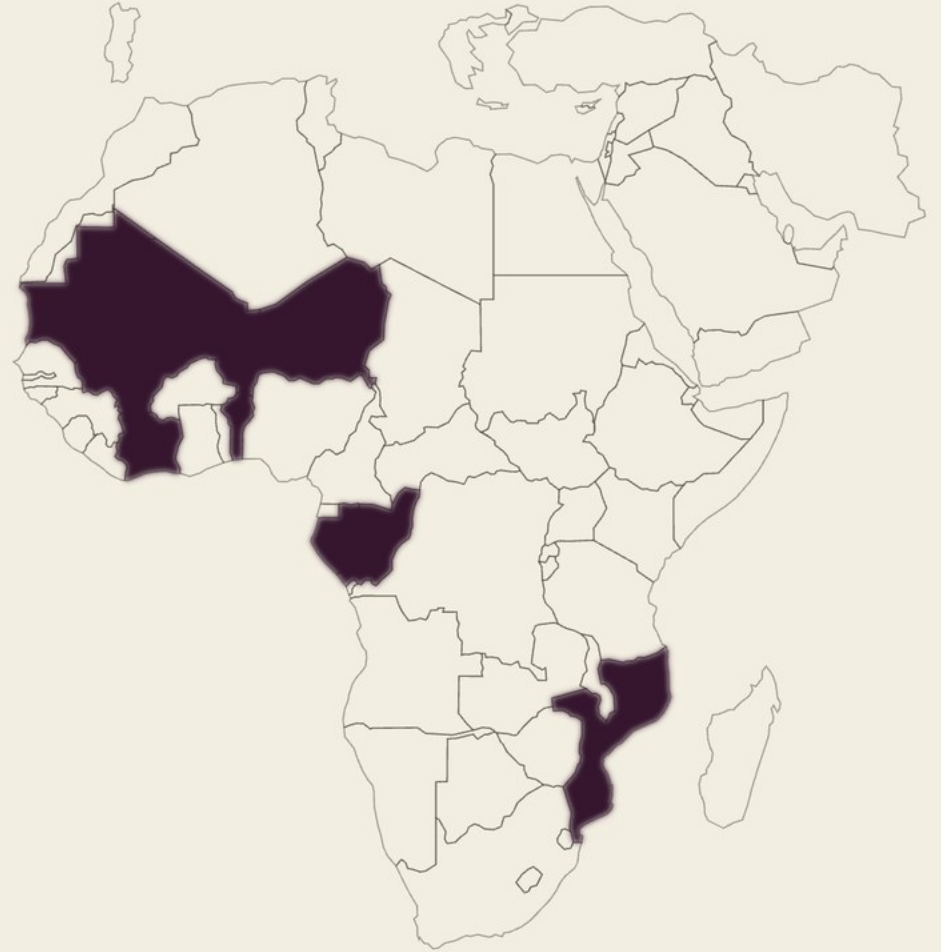


PRIMARY PEA · WOMEN IN DATA DATATHON 2026

Ink and Iron

Do policy records predict maternal anaemia in sub-Saharan Africa?



Similar income. More than three times the anaemia.

Rwanda

GDP per person \$994 (2023)

16%

of women 15–49 anaemic, 2023

Mali

GDP per person \$893 (2023)

57%

of women 15–49 anaemic, 2023



Across 49 countries, income
explains **under 5%** of the difference.
We asked what explains the rest.

Primary Pea

Track: Eat.



Hope Winsor

BS Data Science
(statistics), Colorado State

World Bank income and
FIES food-insecurity data



Alison Nichols

MS Business Analytics,
Santa Clara

UNICEF women's nutrition
EDA and the concern
score



Tatiana Gabel

BS Computer Science +
Data Science, Colorado
State

GIFNA policy feature table



Ramyashree Shetty

Data engineer, Radix,
Mumbai

Presentation lead and
timeline



Saarah Hossain

MEng Software
Engineering, Western
University

The country panel, the
models and the tests

THE PROBLEM

The last mile of the food system is measured in **blood**.

Anaemia in women 15–49 feeds straight into low birthweight and maternal risk. 49 countries, 2000 to 2023.

2023

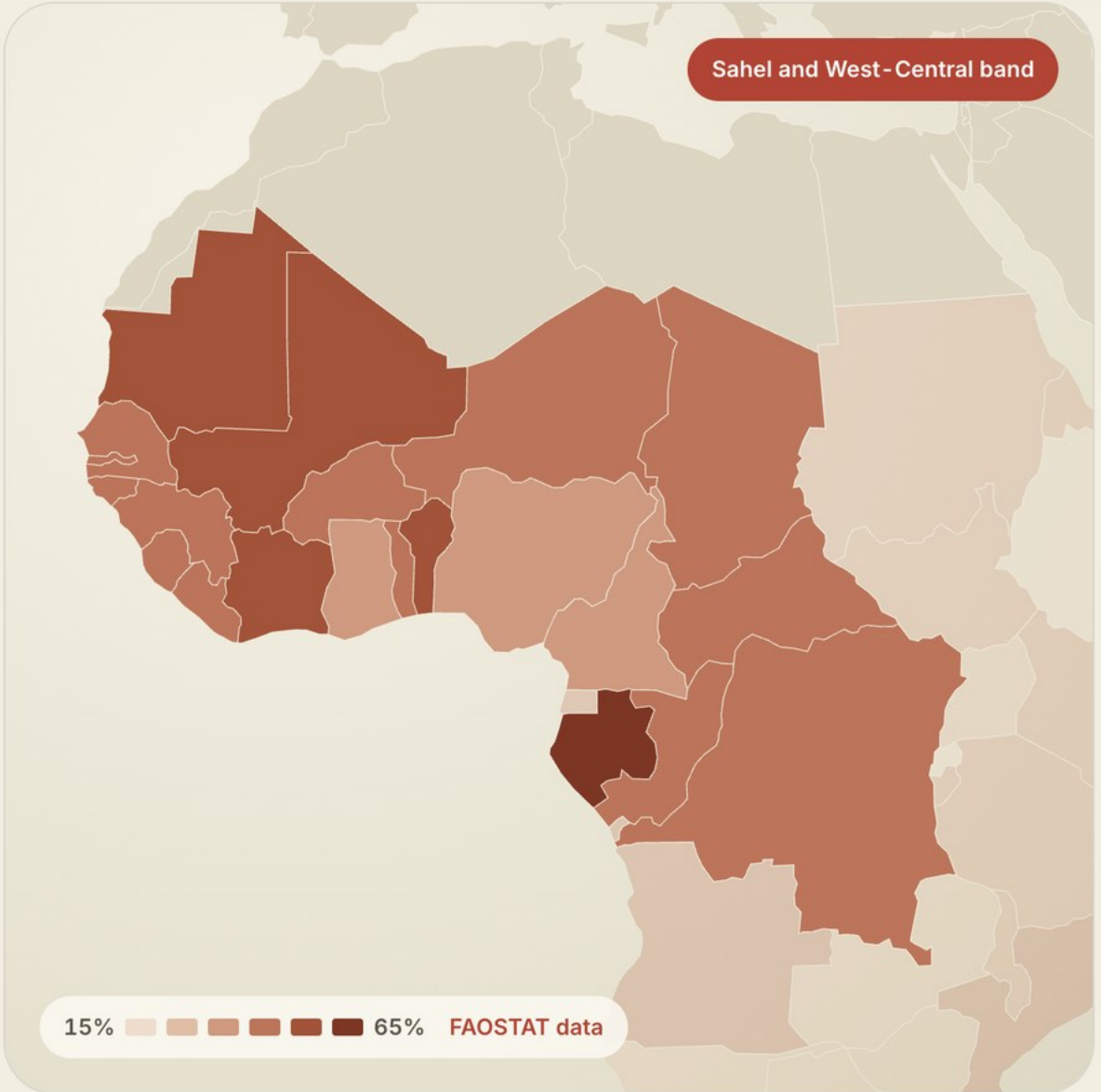
36.7%

REGIONAL MEAN

59.9%

HIGHEST — GABON

FAOSTAT Food Security and Nutrition, items 21043 and 210430, July 2026 release. Regional prevalence is population-weighted across the 49 UNICEF-defined countries; <0.1 million affected-women values are estimated as 0.05 million.



OUR QUESTION

Does a country's policy record predict its maternal anaemia rate?

Sub-Saharan Africa, 2000 to 2023. Hold income and shared shocks fixed, then ask three things: where anaemia is high and which way it is moving, whether a country's policy record moves with its anaemia, and who beats what their context predicts.



QUESTION 1

Where is anaemia high, and which way is it moving?

QUESTION 2

Does the policy record move with anaemia?

QUESTION 3

Who beats their context?

DATA

A dozen datasets, eight organisations.

Each dataset was pulled once, from one release, and keyed to a country code. Names are never keys. Nothing is imputed.

Releases: FAOSTAT July 2026, CAHD 7S2026, UN IGME 2025, UNICEF August 2025, World Bank WDI and DHS API pulls of September 2026, WHO PCT databank November 2025, Nunn & Puga 2012. Two releases of one series never share an axis.



UNICEF · Women's nutrition
Anaemia, underweight and overweight in women



FAOSTAT · Food security
Anaemia, low birthweight, child growth



FAOSTAT · Cost of a healthy diet
Cost and unaffordability, 2017 onward



FAOSTAT · FIES
Food insecurity in 3-year windows



UN IGME · Child mortality
Neonatal, infant and under-5



World Bank · Income and context
GDP per capita in constant 2015 US\$; malaria, fertility, urbanisation, sanitation



WHO GIFNA · Policy registry
Three registry files read directly; dated policies are the dose



GFDx · Fortification laws
Mandates and the year they passed



DHS · Survey rounds
The years a haemoglobin survey anchors the estimate



WHO PCT databank · Deworming
Delivered treatment coverage, school-age children

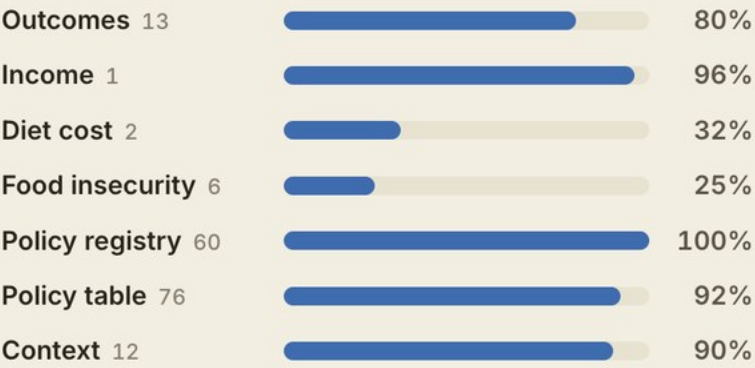
DATA

One table everyone reads.

One guarded script joins the sources. It refuses duplicate keys, undeclared gaps and an incomplete grid. One command rebuilds it; a Colab notebook re-fits the headline model to the same number; 24 automated tests check every build. All of it is public at [primary-pea.github.io](https://github.com/primary/pea).

49 26 174

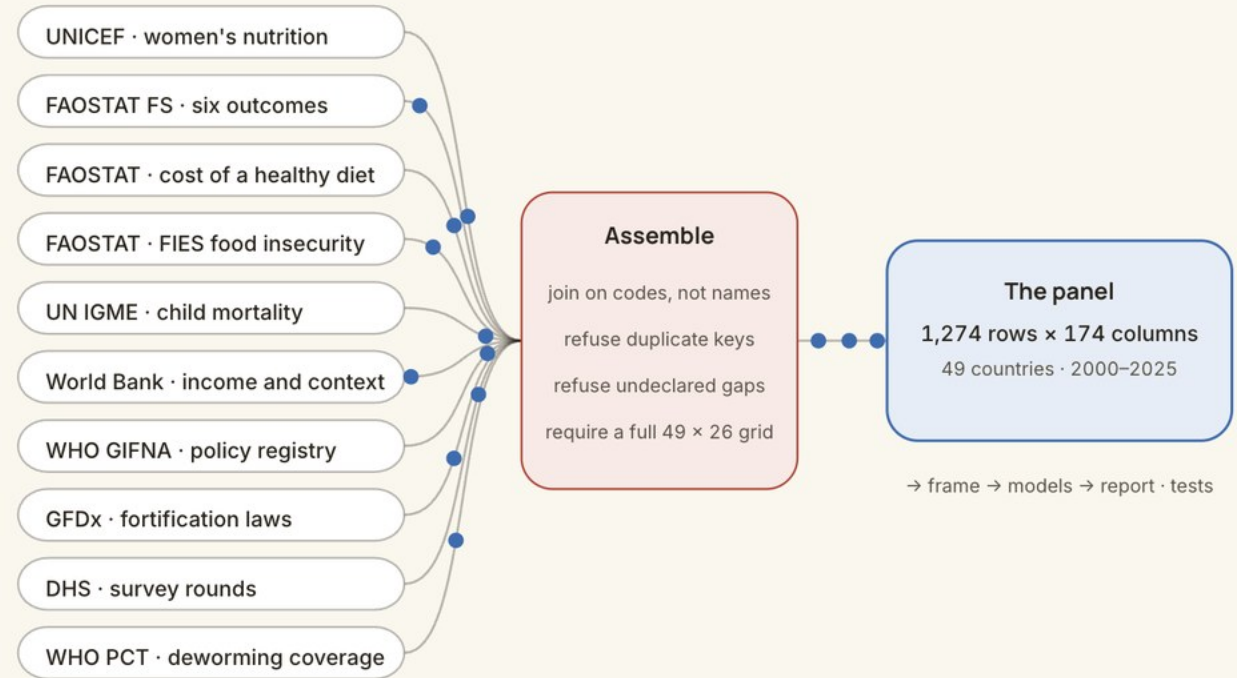
COUNTRIES YEARS COLUMNS



Share of cells filled. Diet cost and food insecurity begin in 2017 and 2014, so they are sparse by design.

From a dozen datasets to one panel

Each source is read once, keyed to a code, and passed through one guarded assembly step.



Each country compared with itself.

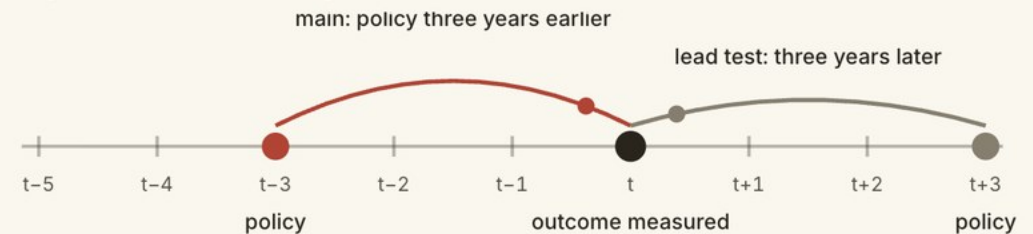
$$\text{anaemia}_{i,t} = a_i + g_t + b \cdot \text{policy}_{i,t-3} + c \cdot \log \text{GDP}_{i,t} + e_{i,t}$$

a_i Country effect. Removes each country's level: geography, health system, malaria. What is left is the country against itself.

g_t Year effect. Removes what hit all 49 at once: price spikes, COVID, changes in WHO's estimates.

Why the lag, and what the lead test checks

Policy three years earlier may predict today's anaemia. Policy three years later cannot; if it "does", governments are reacting to a trend.



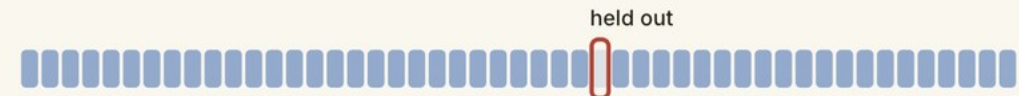
If the lead "predicts", governments are reacting to a trend, not policies driving one. It does: -0.34 vs -0.21 .

b The number we want. Within a country, how anaemia moves after more anaemia policy is in force three years earlier.

c Income control, in logs, so a doubling counts the same at \$800 as at \$8,000.

Predict a country the model has never seen

Fit on 48 countries, predict the 49th's whole path, repeat 49 times. Compare the error with and without policy.



fit on the other 48 · country + year effects, lagged policy, log income

predict its whole 2000–2023 path · repeat × 49 · pool the errors

Result: policy lowers a country's own error in 22 of 49, chance ($p = 0.57$). Error $1.92 \rightarrow 1.93$ points.

Each country compared with itself.

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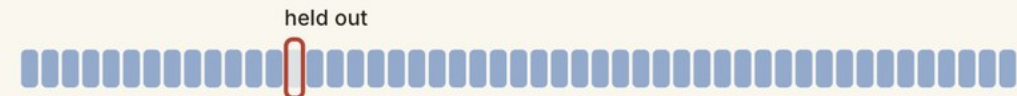
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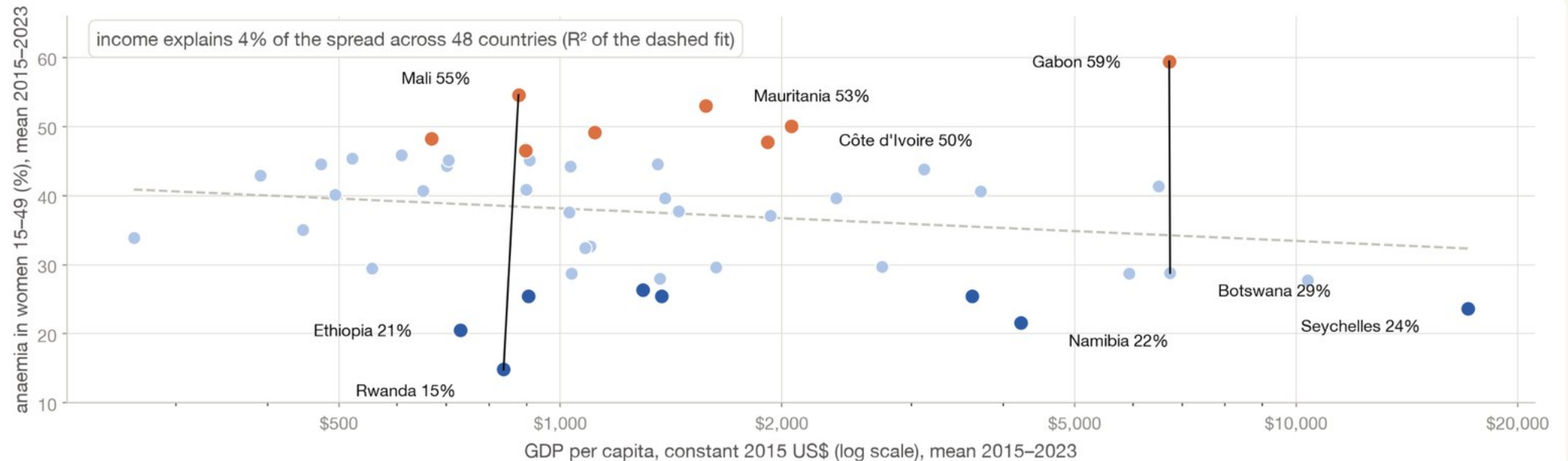
Result: policy lowers a country's own error in 22 of 49, chance ($p = 0.57$). Error $1.92 \rightarrow 1.93$ points.

and money is not the answer.

[Explore](#)[Present](#)

We scored 194 countries on four indicators: women of child bearing age anemia, pregnant women anemia, women's underweight proportion, and women's overweight proportion, looking at both the direction these indicators moved and the country's percentile rank compared to other countries. The five of most concern are Burundi, Jordan, Somalia, Lesotho and Lebanon. We focused down into Sub Saharan Africa where 3 of the 5 countries were located. Averaged over the 49 countries, anaemia fell from 44% to 37% between 2000 and 2015, then turned: higher in 2023 than in 2015 in **37 of 49** countries. Income alone explains **under 5%** of the differences between countries.

Anaemia against income, 2015–2023 means. Orange: the eight highest-anaemia countries. Dark blue: the eight lowest. Vertical lines join two same-income pairs.

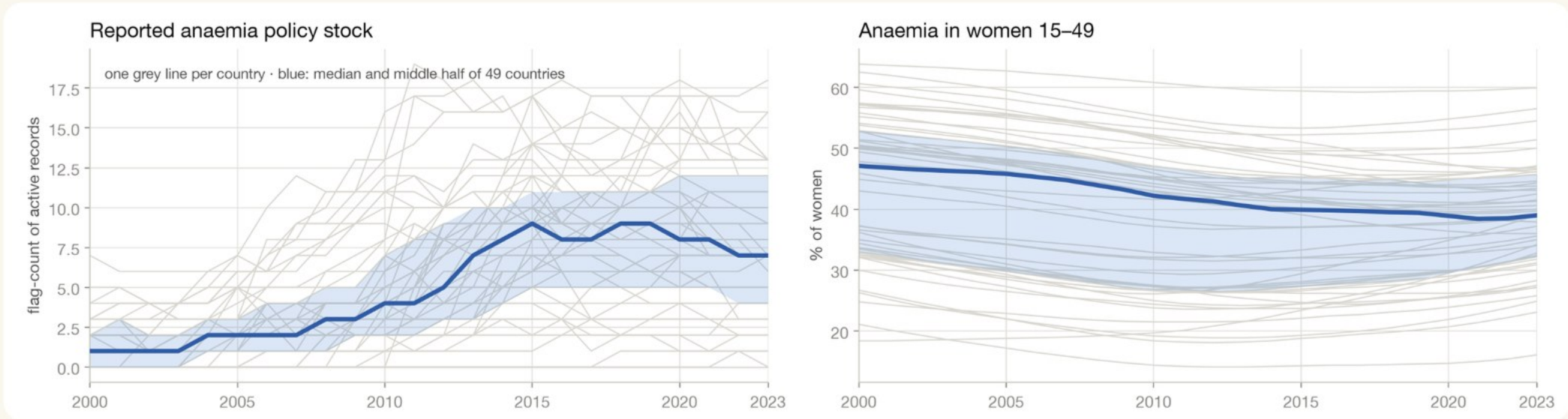


FINDING 2

Both lines move together.

Policy stock rises and anaemia drifts down in nearly every country. A naive model calls that an effect. **The test is to separate the policy from the shared trend.**

Reported anaemia policy stock (left, flag-count of active records) and anaemia in women 15–49 (right, %), 2000–2023

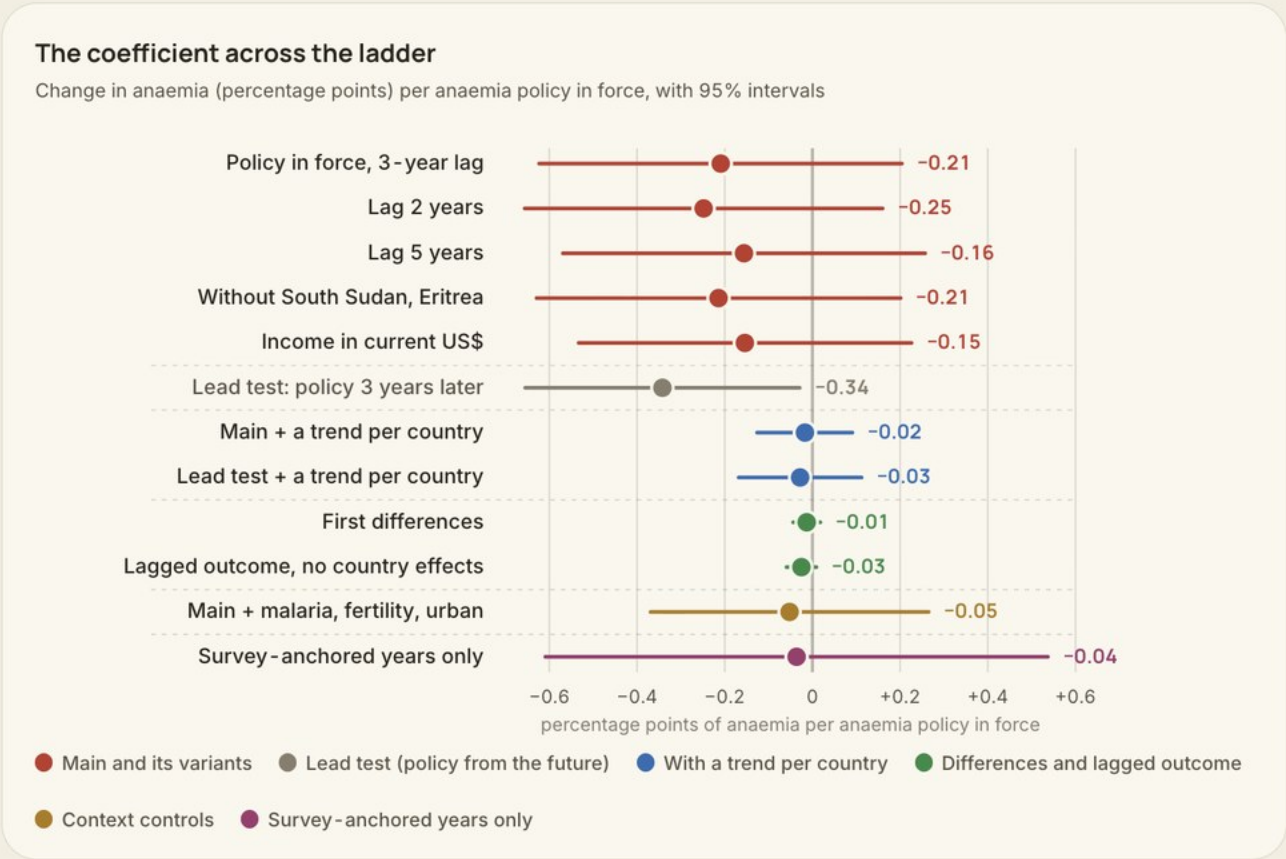


One grey line per country; blue is the median and middle half of 49 countries. WHO anaemia via FAOSTAT; WHO GIFNA registry. Scaffold v2.0, data afe63e46c26b.

FINDING 2

The policy record does not move anaemia, measurably.

Inside a country, one more anaemia policy in force is followed three years later by **-0.2 percentage points, indistinguishable from zero**. The one row off zero is the lead: policies from the future "predict" today's anaemia, the fingerprint of governments reacting to a trend.



In numbers

Scaffold v2.0 results, every value tagged in RESULTS.md

- 

-0.21

Main model. Anaemia policies in force three years earlier; the interval spans zero (-0.62 to +0.20).
- 

-0.34

Lead test. Policy from the future, $p = 0.03$: governments reacting to trends, not policies driving them.
- 

+0.10 / -0.08

Around each country's first policy: two-stage DiD and local-projections DiD sit at zero. No robust break.
- 

≥ 0.6 pp

Benefits of six tenths of a point per policy or more are ruled out; smaller effects could not be detected (336 specifications, 7% exclude zero).

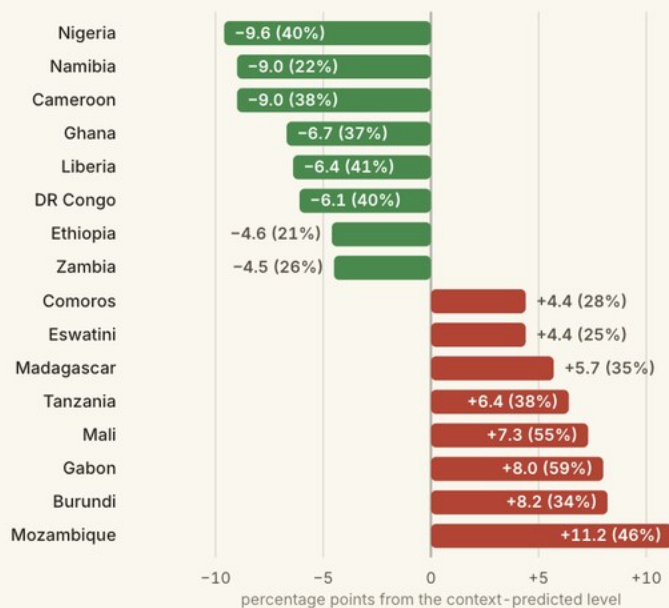
Who beats their context, and the paper trail cannot say.

[Explore](#)[Present](#)

Income, fertility, malaria, urbanisation, sanitation, sub-region and terrain explain 81% of the differences in levels. Nigeria, Namibia, Cameroon, Ghana and Liberia sit six to ten points below the level their context predicts. On the books, their policy record is **not clearly different** from the eight above them.

Anaemia relative to what context predicts

Percentage points above or below the context model, mean 2015–2023, level in brackets



● Lower than context predicts ● Higher than context predicts

What the two groups have on the books

Mean reported policy stock, 2015–2023

Anaemia policies in force (dated) 4.5 2.8 p = 0.12

Anaemia records, all three files 5.4 3.7 p = 0.17

Anaemia programming (team composite) 10.8 6.6 p = 0.07

Active policies, any topic 15.7 11.0 p = 0.13

● Lower eight ● Higher eight

RECOMMENDATIONS

Measure delivery, not documents.



DATA CUSTODIANS · WHO, FAO

Record dates and coverage for every programme

Three in four programme records carry no date, and the coverage field is empty. Delivery data is the missing link.



MINISTRIES · HIGH-BURDEN BAND

Report coverage at the point of delivery

Supplementation and fortification reach, not policy counts. Learn from the countries below their context: Nigeria, Namibia, Cameroon, Ghana, Liberia.



FUNDERS AND RESEARCHERS

Anchor the estimates, pre-register the effect worth finding

More surveys that measure haemoglobin (33 of 49 countries have one in the window), and a smallest effect of interest set in advance, so a null reads as a bound, not an absence.

Next for us: subnational data, cost-of-diet linkage, sensitivity bounds for the event study, and schistosomiasis treatment coverage.

The last mile is measured in **blood**, not in documents.

Until delivery is recorded, nobody can say which policies work.



record what reaches women, not what is written

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Data: WHO, UNICEF, FAOSTAT, UN IGME, World Bank, WHO GIFNA, GFDx, DHS, WHO PCT databank, Nunn & Puga. Illustrations: Storyset, unDraw. Icons: Microsoft Fluent Emoji. Scaffold v2.0, data version afe63e46c26b.

